Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A sample of hard water contains $1 \mathrm{mg} \mathrm{CaCl}_2$ and $1 \mathrm{mg} \mathrm{MgCl}_2$, per litre. Calculate the hardness of water in terms of CaCO_3 present in per 10^6 parts of water.

(A) $2.5 \mathrm{ppm}$

(B) $1.95 \mathrm{ppm}$

(C) $2.15 \mathrm{ppm}$

(D) $195 \mathrm{ppm}$

Q.2

Hardness of water is $200 \mathrm{ppm}$. The normality and molarity of CaCO_3 in the water is

(A) $2 \times 10^{-6} \mathrm{~N}, 2 \times 10^{-6} \mathrm{M}$

(B) $4 \times 10^{-2} \mathrm{~N}, 2 \times 10^{-2} \mathrm{M}$

(C) $4 \times 10^{-3} \mathrm{~N}, 2 \times 10^{-3} \mathrm{M}$

(D) $4 \times 10^{-1} \mathrm{~N}, 2 \times 10^{-1} \mathrm{M}$

Q.3

A sample of hard water contains 122 ppm of $\mathrm{HCO}_3^{\ominus}$ ions. What is the minimum weight of CaO required to remove ions completely from $1 \mathrm{~kg}$ of such water sample?

(A) $56 \mathrm{mg}$

(B) $112 \mathrm{mg}$

(C) $168 \mathrm{mg}$

(D) $244 \mathrm{mg}$

Q.4

$100 \mathrm{~mL}$ samples of distilled water, tap water and boiled water required, respectively, $2 \mathrm{~mL}, 17 \mathrm{~mL}$ and $7 \mathrm{~mL}$ of soap solution to form permanent lather. The ratio of permanent to temporary hardness in the tap water is

(A) $3:2$

(B) $2:3$

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(C) $1:2$

(D) $2:1$

Q.5

$3.4\ \text{g}$ sample of H_2O_2 solution containing $x\%H_2O_2$ by weight requires $x\ \text{mL}$ of a $KMnO_4$ solution for complete oxidation under acidic condition. The normality of $KMnO_4$ solution is:

(A) $1\ \text{N}$

(B) $2\ \text{N}$

(C) $3\ \text{N}$

(D) $0.5\ \text{N}$

Q.6

If $100\ \text{mL}$ of acidified $2\ \text{N}H_2O_2$ is allowed to react with $KMnO_4$ solution till there is light tinge of purple colour, the volume of oxygen produced at STP is:

(A) $2.24\ \text{L}$

(B) $1.12\ \text{L}$

(C) $3.36\ \text{L}$

(D) $4.48\ \text{L}$

Q.7

A $5.0\ \text{mL}$ of solution of H_2O_2 liberates $0.508\ \text{g}$ of iodine from acidified KI solution. Calculate the strength of H_2O_2 solution in terms of volume strength at STP.

(A) 4.48

(B) 1.34

(C) 3.56

(D) 2.73

Q.8

To a $25\ \text{mL}H_2O_2$ solution, excess of an acidified solution of potassium iodine was added. The iodine liberated required 20 mL of $0.3\ \text{N}$ sodium thiosulphate solution. Calculate the volume strength of H_2O_2 solution.

(A) 4.48

(B) 1.34

(C) 3.56

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(D) 2.73

Q.9

Element (A) burns in nitrogen to give an ionic compound, (B) reacts with water to give (C) and (D). A solution of (C) becomes milky on bubbling carbon dioxide. Identify (D)

(A) Ca_3N_2

(B) $Ca(NO_3)_2$

(C) NH_3

(D) None

Q.10

In which of the following reactions H_2O_2 acts as a reducing agent?
(i) $H_2O_2 + 2H^+ + 2e^- \rightarrow 2H_2O$
(ii) $H_2O_2 - 2e^- \rightarrow O_2 + 2H^+$
(iii) $H_2O_2 + 2e^- \rightarrow 2OH^-$
(iv) $H_2O_2 + 2OH^- - 2e^- \rightarrow O_2 + 2H_2O$

(A) (ii) and (iv)

(B) (i) and (ii)

(C) (iii) and (iv)

(D) (i) and (iii)

Q.11

Hydrogen peroxide acts both as an oxidising and as a reducing agent depending upon the nature of the reacting species. In which of the following cases H_2O_2 acts as a reducing agent in acid medium?

(A) MnO_4^-

(B) $Cr_2O_7^{2-}$

(C) SO_3^{2-}

(D) KI

Q.12

Permanent hardness in water cannot be cured by

(A) boiling

(B) ion exchange method

(C) Calgon' method

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(D) treatment with washing soda

Q.13

The combination of hydrogen and carbon monoxide in the presence of the catalyst ZnO and Cu gives

(A) $\mathrm{C}+\mathrm{H}_2\mathrm{O}$

(B) $\mathrm{CH}_4+\mathrm{H}_2\mathrm{O}$

(C) HCHO

(D) $\mathrm{CH}_3\mathrm{OH}$

Q.14

Which of the following orders of boiling points of hydrides of Group 16 is correct?

(A) $\mathrm{H}_2\mathrm{O}>\mathrm{H}_2\ \mathrm{S}>\mathrm{H}_2\mathrm{Se}$

(B) $\mathrm{H}_2\mathrm{O}>\mathrm{H}_2\ \mathrm{S}<\mathrm{H}_2\mathrm{Se}$

(C) $\mathrm{H}_2\mathrm{O}<\mathrm{H}_2\ \mathrm{S}>\mathrm{H}_2\mathrm{Se}$

(D) $\mathrm{H}_2\mathrm{O}<\mathrm{H}_2\ \mathrm{S}<\mathrm{H}_2\mathrm{Se}$

Q.15

When $\mathrm{H}_2\mathrm{O}$ is oxidized the product is

(A) O^{2-}

(B) OH^{-}

(C) HO_2^{-}

(D) O_2

Q.16

Which of the following equations depicts the oxidizing nature of $\mathrm{H}_2\mathrm{O}_2$?

(A) $2\mathrm{MnO}_4^{-}+6\mathrm{H}^{+}+\mathrm{H}_2\mathrm{O}_2\rightarrow 2\mathrm{Mn}^{2+}+8\mathrm{H}_2\mathrm{O}+5\mathrm{O}_2$

(B) $2\mathrm{Fe}^{3+}+2\mathrm{OH}^{-}+\mathrm{H}_2\mathrm{O}_2\rightarrow 2\mathrm{Fe}^{2+}+2\mathrm{H}_2\mathrm{O}+\mathrm{O}_2$

(C) $2\mathrm{I}^{-}+2\mathrm{H}^{+}+\mathrm{H}_2\mathrm{O}_2\rightarrow \mathrm{I}_2+2\mathrm{H}_2\mathrm{O}$

(D) $\mathrm{KIO}_4+\mathrm{H}_2\mathrm{O}_2\rightarrow \mathrm{KIO}_3+\mathrm{H}_2\mathrm{O}+\mathrm{O}_2$

Q.17

The normality of ' 30 volume $\mathrm{H}_2\mathrm{O}_2$ ' is

(A) $2.678\ \mathrm{N}$

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(B) 5.336 N

(C) 8.034 N

(D) 6.685 N

Q.18

20 mL of a sample of H_2O_2 gives 400 mL of O_2 measured at STP. The sample should be labelled as

(A) 5 volume H_2O_2

(B) 10 volume H_2O_2

(C) anhydrous H_2O_2

(D) 20 volume H_2O_2

Q.19

In the reaction

(i) $H_2O_2 + O_3 \rightarrow 2H_2O + 2O_2$

(ii) $PbS + 4O_3 \rightarrow PbSO_4 + 4O_2$

(A) O_3 is reduced both in (i) and (ii)

(B) O_3 is oxidized both in (i) and (ii)

(C) O_3 is oxidized in (i) and reduced in (ii)

(D) O_3 is reduced in (i) and oxidized in (ii)

Q.20

Which of the following substance in the fused/liquid state will liberate H_2 at anode on electrolysis?

(A) $NaOH$

(B) CaH_2

(C) HCl

(D) H_2O

Q.21

10ml of H_2O_2 liberates 12.7 g of iodine from an acidic KI solution. Calculate normality

Q.22

Calculate the volume of 10 volume H_2O_2 solution that will react with 200 mL of 2 N$KMnO_4$ in acidic medium.

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Q.23

Calculate weight of O_2 produced on heating $1.12\ L$ of 10 volume strength of H_2O_2 at STP.

Q.24

Calculate percentage strength of 10 volume H_2O_2.

Q.25

A water sample is found to contain $96ppm$ of SO_4^{2-} and 122 ppri of HCO_3^{Θ} with Ca^{2+} ion as the only cation. Calculate the mol of CaO required to remove HCO_3^{Θ} ion from $1000\ kg$ of the water.

Answers & Solutions

Q.1 Answer:

$1.95ppm$

Solution:

Molecular weight of $CaCl_2 = 111.0\ g$

MW **of** $CaCO_3 = 100\ g$

MW **of** $MgCl_2 = 95.0\ g$

$CaCl_2 + Na_2CO_3 \longrightarrow CaCO_3 + 2NaCl$

$MgCl_2 + Na_2CO_3 \longrightarrow MgCO_3 + 2NaCl$

(i) $111.0\ gCaCl_2 \equiv 100\ gCaCO_3$

$1mgCaCl_2 \equiv \frac{100}{111}mgCaCO_3 = 0.9mgCaCO_3$

(ii) 95.0 $gMgCl_2 \equiv 100\ gCaCO_3$

$1mgMgCl_2 \equiv \frac{100}{95}mgCaCO_3 = 1.05mgCaCO_3$

Hardness of $CaCO_3ppm$

$1mgMgCl_2 \equiv \frac{100}{95}mgCaCO_3 = 1.05mgCaCO_3$

$\equiv \frac{(0.9+1.05)\times10^{-3}\ g\times10^6\ mL}{10^3\ mL}$

$= 1.95ppm$

Q.2 Answer:

$4 \times 10^{-3}\ N, 2 \times 10^{-3}M$

Solution:

Hardness is weight in grams of $CaCO_3$ **in** $10^6\ mL$ **of** H_2O **or** $= 10^6\ g$ **of** H_2O

$(\because d H_2O = 1)$

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$\mathrm{MW}\left(\mathrm{CaCO}_3\right)=100 \mathrm{~g} \mathrm{~mol}^{-1}$

$M=\frac{W_2 \times 1000}{M W \times \text { Volume of solution in mL }}$

$=\frac{200 \times 1000}{100 \times 10^6}=2 \times 10^{-3} \mathrm{M}$

$N=n \times M=2 \times 2 \times 10^{-3}=4 \times 10^{-3} \mathrm{~N}$

Hence, the correct option is (c).

Q.3 Answer:

56mg

Solution:

$\left[\mathrm{MW}\left(\mathrm{HCO}_3^{\ominus}\right)=61 \mathrm{~g}, \mathrm{MW}(\mathrm{CaO})=56 \mathrm{~g}\right]$

Temperary hardness is due to the $\mathrm{HCO}_3^{\ominus}$ of Ca^{2+} and Mg^{2+} and is removed by either boiling or by treating with CaO or $\mathrm{Ca}(\mathrm{OH})_2$ (Clark's method).

$$\underset{1 \mathrm{mmol}}{\mathrm{Ca}\left(\mathrm{HCO}_3\right)_2}+\underset{1 \mathrm{mmol}}{\mathrm{CaO}} \longrightarrow \underset{2 \mathrm{mmol}}{2 \mathrm{CaCO}_3}+\mathrm{H}_2 \mathrm{O}$$

$122 \mathrm{ppm}$ **of** $\mathrm{HCO}_3^{\ominus} \equiv 122 \mathrm{~g} \mathrm{HCO}_3^{\ominus}$ **in** $10^6 \mathrm{~mL}$ **or** $10^3 \mathrm{~L}$

$\equiv 122 \mathrm{mg} \mathrm{HCO}_3{ }^{\ominus}$ in 1.0 L

$\equiv \frac{122}{61}=2 \mathrm{mmol}$ in 1.0 L

According to Eq. (i),

$2 \mathrm{mmol}$ **of** $\mathrm{Ca}\left(\mathrm{HCO}_3\right)_2 \equiv 2 \mathrm{mmol}$ **of** CaO **in** 1.0 L

$1 \mathrm{mmol}$ **of** $\mathrm{Ca}\left(\mathrm{HCO}_3\right)_2 \equiv 1 \mathrm{mmol}$ **of** CaO **in** 1.0 L

$\equiv 1 \times 56 \equiv 56 \mathrm{mg}$ of CaO

$\equiv 56 \mathrm{mg}$ of CaO in 1.0 L

$\Rightarrow 56 \mathrm{mg}$ of CaO in 1 kg of $\mathrm{H}_2 \mathrm{O}$

$(1 \mathrm{~L}$ **of** $\mathrm{H}_2 \mathrm{O}=1 \mathrm{~kg}$ **of** $\mathrm{H}_2 \mathrm{O}, \because d \mathrm{H}_2 \mathrm{O}=1)$

Hence, the correct option is (a).

Q.4 Answer:

1 : 2

Solution:

\begin{array}{|l|c|c|}

**\hline & Distilled \mathbf{H}_{2} \mathrm{O} (No hardness, lather factor) & Tap \mathbf{H}_{2} \mathbf{O} (Temporary + permanent hardness) & Boiled \mathbf{H}_{2} \mathbf{O} (Only permanent hardness; temporary hardness is removed by boiling) **

**\hline Volume of soap solution required & 2 \mathrm{~mL} & 17 \mathrm{~mL} & 7 \mathrm{~mL} **

**\hline Volume of soap effectively used & - & 17-2 =15 \mathrm{~mL} & 7-2=5 \mathrm{~mL} **

\hline

\end{array}

(Temporary and permanent hardness) in tap $\mathrm{H}_2 \mathrm{O}=15 \mathrm{~mL}$

Permanent hardenss in boiled $\mathrm{H}_2 \mathrm{O}=5 \mathrm{~mL}$

Temporary hardness in $\mathrm{H}_2 \mathrm{O}=15-5=10 \mathrm{~mL}$

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$\frac{\text{Permanent hardness}}{\text{Temporary hardness}} = \frac{5}{10} = 1:2$

Hence, answer is (c).

Q.5 Answer:

$2\ \mathrm{N}$

Solution:

$100\ \mathrm{g}$ **of** $\mathrm{H_2O_2}$ **sample contains** $= 'x'\mathrm{g H_2O_2}$ $3.4\ \mathrm{g}$ **of** $\mathrm{H_2O_2}$ **sample solution contains** $= \frac{x}{100} \times 3.4\ \mathrm{g}$

Weight of $\mathrm{H_2O_2} = \frac{3.4x}{100}$

Eq of $\mathrm{H_2O_2} = \frac{3.4x}{100} \times \frac{1}{17}$

mEq **of** $\mathrm{H_2O_2} = \frac{3.4x}{100 \times 17} \times 1000 = \frac{34x}{17} = 2x$

mEq **of** $\mathrm{KMnO_4} = x \times \mathrm{N}$

$\therefore \quad x \times N = 2x$

$\mathrm{N} = 2$

Hence, the correct option is (b).

Q.6 Answer:

$2.24\ \mathrm{L}$

Solution:

Volume of $\mathrm{O_2}$ **at** $\mathrm{STP} = 100\ \mathrm{mL} \times 11.2$ **volume strength** $= 1120\ \mathrm{mL}$ **of** $\mathrm{O_2}$ **at STP**

(Since $1N = 5.6$ **volume strength;** $2N = 11.2$ **volume strength of** $\mathrm{H_2O_2}$

Volume of $\mathrm{O_2}$ **produced by** $\mathrm{H_2O_2} = 1120\ \mathrm{mL}$

Same volume of $\mathrm{O_2}$ **produced by** $\mathrm{KMnO_4} = 1120\ \mathrm{mL}$

Total volume of $\mathrm{O_2} = 2240\ \mathrm{mL} = 2.24\ \mathrm{L}$ **Hence, the correct option is (a).**

Q.7 Answer:

4.48

Solution:

mEq **of** $I_2 = \frac{0.508}{127} \times 1000$

mEq **of** $\mathrm{KI} = \frac{0.508}{127} \times 1000$

mEq **of** $\mathrm{H_2O_2} = \frac{0.508}{127} \times 1000$

Normality of $\mathrm{H_2O_2} = \frac{0.503}{127} \times \frac{1000}{5}$

For $\mathrm{H_2O_2}$

$5.6\times$ **Normality** $=$ **Volume strength** $\therefore$ **Volume strength** $= 5.6 \times \frac{0.508 \times 1000}{127 \times 5} = 4.48\ \mathrm{V}$

Q.8 Answer:

1.34

Solution:

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$\text{mEq of } Na_2S_2O_3 - 20 \times 0.3 = 6$

$\text{mEq of } I_2 = 6$

$\text{mEq of } H_2O_2 = 6$

Normality of $H_2O_2 = \frac{6}{25}$

For $H_2O_2, 5.6\times$ **Normality = Volume strength**

$\therefore$ **Volume strength** $= 5.6 \times \frac{6}{25} = 1.344$

Q.9 Answer:

NH_3

Solution:

a. Since element (A) burns in nitrogen to give an ionic compound, therefore (B) must be a metal nitride

b. Since (B), a metal nitride reacts with water, (B) is ionic nitride and the product formed, i.e. (C) and (D) are metal hydroxides and ammonia, (NH_3)**.**

c. Since (C) becomes milky on bubbling CO_2**,** (C) **must by calcium hydroxide,** $Ca(OH)_2$**.**

All reactions involved in the question can be explained as follows:

$$3Ca_{(s)} + N_{2(g)} \xrightarrow{\Delta} Ca_3N_{2\,(s)}$$

Calcium nitride

(A) (B)

$$Ca_3N_2 + 6H_2O \longrightarrow Ca(OH)_2 + 2NH_{3\,(g)}$$

(B) Calcium hydroxide(C) Ammonia (D)

$$Ca(OH)_2 + CO_2 \longrightarrow CaCO_3\downarrow + H_2O$$

Calcium carbonate (milkiness)

(D)

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Thus, $(A)=Ca,(B)=Ca_3N_2,(C)=Ca(OH)_2(D)=NH_3$

Q.10 Answer:

(ii) and (iv)

Solution:

The reactions (ii) and (iv) releases electrons for the reduction of other species. Thus, H_2O_2 **in reactions (ii) and (iv) acts as a reducing reagent.**

Q.11 Answer:

MnO_4^-

Solution:

With MnO_4^- **in acidic medium,** H_2O_2 **acts as a reducing agent.**

$2MnO_4^- + 6H^+ + 5H_2O_2 \rightarrow 2Mn^{2+} + 8H_2O + 5O_2$

H_2O_2 **oxidizes** SO_3^{2-} **to** SO_4^{2-}, I^- **to** I_2 **and** $Cr_2O_7^{2-}$ **to** CrO_3 **in acidic medium.**

$SO_3^{2-} + H_2O_2 \rightarrow SO_4^{2-} + H_2O$

$2I^- + 2H^+ + H_2O_2 \rightarrow 2H_2O + I_2$

$Cr_2O_7^{2-} + 2H^+ 4H_2O_2 \rightarrow \underset{\text{(perchromate)}}{2CrO_5} + 5H_2O$

Q.12 Answer:

boiling

Solution:

Permanent hardness in water cannot be cured by boiling the water.

Q.13 Answer:

CH_3OH

Solution:

$CO + 2H_2 \xrightarrow{ZnO/Cu} CH_3OH$

Q.14 Answer:

$H_2O > H_2S < H_2Se$

Solution:

Because of hydrogen bondings, the boiling point of H_2O **is greater than that of** H_2S**. The boiling point of** H_2S **is smaller than** H_2Se **as the former is a lighter molecule.**

Q.15 Answer:

O_2

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Solution:

The oxidation of H_2O is $2H_2O \rightarrow 4H^+ + 4e^- + O_2$

Q.16 Answer:

$2I^- + 2H^+ + H_2O_2 \rightarrow I_2 + 2H_2O$

Solution:

In the oxidizing nature of H_2O_2, the oxidation number of O atom is changed from -1 to -2.

Q.17 Answer:

5.336 N

Solution:

1 L of ' 30 volume H_2O_2 ' liberates 30 L of O_2 at STP or $(30/22.4)$mol of O_2. From the equation

$2H_2O_2 \rightarrow H_2O + O_2$

it is obvious that $(30/22.4)$mol of O_2 is liberated from $(60/22.4)$mol of H_2O_2, i.e. 2.678 mol. Hence, molarity of H_2O_2 is 2.678 mol L^{-1} and its normality will be 2×2.678 eq L^{-1}.

Q.18 Answer:

20 volume H_2O_2

Solution:

20 mL of H_2O_2 liberates 400 mLO_2 at STP. 1000 mL of H_2O_2 liberates 400×50 mL$(=20$ L$)$ of O_2 at STP Hence, solution is ' 20 volume H_2O_2'

Q.19 Answer:

O_3 is oxidized in (i) and reduced in (ii)

Solution:

Eq. (i) may be represented as

$H_2O_2 \rightarrow H_2O + O$ and $O_3 + O \rightarrow 2O_2$

Eq. (ii) may be represented as

$O_3 \rightarrow O_2 + O$ and $PbS + 4O \rightarrow PbSO_4$

Q.20 Answer:

CaH_2

Solution:

Electrolysis of molten CaH_2 produces hydrogen gas at anode.

In CaH_2, hydrogen is present in the form of hydride $H-$.

Oxidation is loss of electrons and we know oxidation occurs at anode and it is also true that in

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calcium hydride, hydrogen is present as hydride.

So, $H-$ will loss electron and form H_2 gas at the anode.

Q.21 Answer:

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Solution:

$\mathrm{MW}\left(\mathrm{I}_2\right)=2 \times 127=254 \mathrm{~g}$
$\mathrm{EW}\left(\mathrm{I}_2\right)=\frac{254}{2}=127 \mathrm{~g}$
$\left(\because 2 \mathrm{I}^{\omega} \longrightarrow \mathrm{I}_2+2 \mathrm{e}^{\mathrm{G}}, n=2\right)$
$\mathrm{H}_2 \mathrm{O}_2 \equiv \mathrm{KI} \equiv \mathrm{I}_2$
$1 \mathrm{mEq} \equiv 1 \mathrm{mEq} \equiv 1 \mathrm{mEq}$
$\mathrm{N}_1 \mathrm{~V}_1(\mathrm{~mL})=\frac{\text{Weight of } \mathrm{I}_2}{\mathrm{EW}\left(\mathrm{I}_2\right)} \times 10^3(\mathrm{mEq})$
$\mathrm{N}_1 \times 10 \mathrm{~mL}=\frac{12.7 \times 10^3}{127}$
N_1 **of** $\mathrm{H}_2 \mathrm{O}_2 \equiv 10 \mathrm{~N}$

Q.22 Answer:

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Solution:

Normality of 10 volume $\mathrm{H}_2 \mathrm{O}_2=\frac{\text{Volume strength}}{5.6}=\frac{10}{5.6} \mathrm{~N}$
Applying normality equation,
$\mathrm{N}_1 \mathrm{~V}_1=\mathrm{N}_2 \mathrm{~V}_2$
$\left(\mathrm{H}_2 \mathrm{O}_2\right) \quad\left(\mathrm{KMnO}_4\right)$
$\frac{10}{5.6} \times \mathrm{V}_1=2 \times 200$
$\mathrm{V}_1=\frac{2 \times 200 \times 5.6}{10}=224 \mathrm{~mL}$

Q.23 Answer:

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Solution:

Volume of O_2 **at STP**
$=$ volume of $\mathrm{H}_2 \mathrm{O}_2 \times$ volume strength of $\mathrm{H}_2 \mathrm{O}_2$
$=1.12 \mathrm{~L} \times 10=11.2 \mathrm{~L}$
Moles of O_2 **at** $\mathrm{STP}=\frac{11.2 \mathrm{~L}}{22.4 \mathrm{~L}}=0.5 \mathrm{~mol}$
Weight of $\mathrm{O}_2=0.5 \times 32=16.0 \mathrm{~g}$

Q.24 Answer:

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Solution:

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5.6 volume strength of $H_2O_2 = 1.7\%$

10 volume strength of $H_2O_2 = \frac{1.7 \times 10}{5.6} = 3.03\%$

Q.25 Answer:

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Solution:

$\underset{1 \text{ mol}}{Ca} + Ca(HCO_3)_2 \longrightarrow \underset{2 \text{ mol}}{CaCO_3} + H_2O$

$= 2 \text{ mol} HCO_3^{\ominus}$ in 10^6 g H_2O

2 mol of $HCO_3^{\ominus}$ is present in 10^6 mL $(= 10^6 \text{ g}) = 1000$ kg of H_2O.

From Eq. (iii), 1 mol of CaO is required to remove 2 mol $HCO_3^{\ominus}$ present in 1000 kg of H_2O.

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